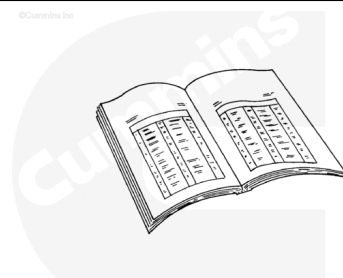


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



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⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

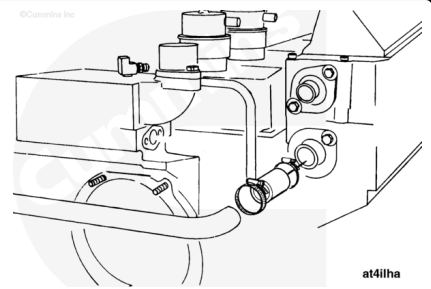
⚠ CAUTION ⚠

Use caution when draining coolant that the coolant is not spilled or drained into the bilge area. Do not pump coolant overboard. If the coolant is not reused, it must be disposed of in accordance with local environmental regulations.

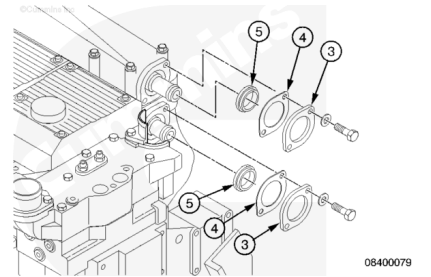
- Drain the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>).
- Remove the air crossover tube. Refer to Procedure 010-019 (</qs3/pubsys2/xml/en/procedures/20/20-010-019-tr.html>).
- Remove the aftercooler coolant vent tube. Refer to Procedure 008-017 (</qs3/pubsys2/xml/en/procedures/20/20-008-017.html>).
- Marine applications: Remove the aftercooler coolant tube. Refer to Procedure 010-006 (</qs3/pubsys2/xml/en/procedures/20/20-010-006.html>).

Remove

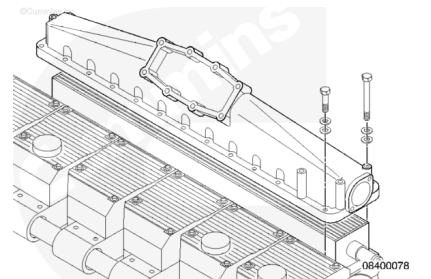
Loosen the two hose clamps and remove the inlet hose.



Remove the capscrews, seal retainer (3), gaskets (4), and seals (5). Discard the seals and gaskets.

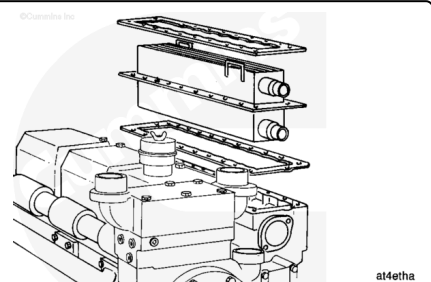


Remove the aftercooler cover capscrews and cover.



Remove the element and gaskets.

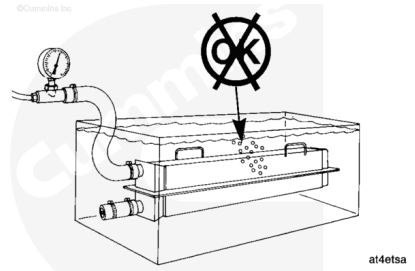
Discard gaskets.



Pressure Test

Use a hose plugged with a pipe plug. Plug one water tube as shown. Attach an air line and gauge to the remaining tube.

Apply air pressure and check for leaks.



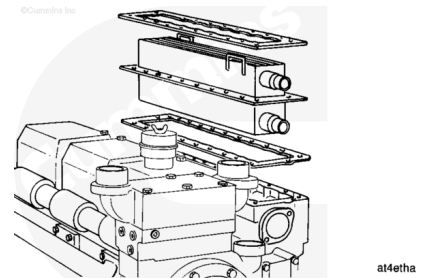
Measurements

	kpa	psi
Air Pressure	415	60

Note : Heating the tank water to 50°C [120°F] improves the test results.

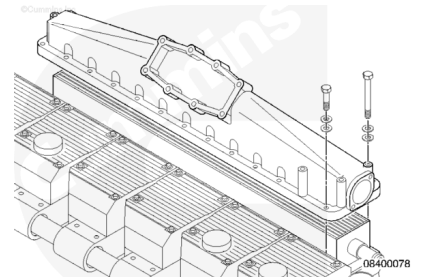
Install

Install the element and two new gaskets.



Install the cover and aftercooler cover capscrews.

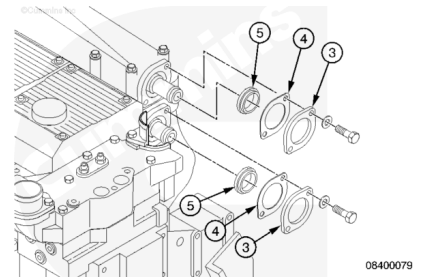
Do **not** tighten the capscrews until the inlet and outlet seals are installed.



Use clean engine oil. Lubricate the seal (5). Install the seal on the retainer (3).

Use clean engine oil. Lubricate the outside diameter of the water bosses on the aftercooler element.

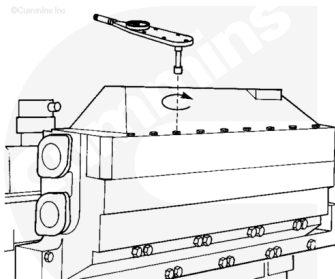
Slide the gasket (4) on the water bosses. Slide the seal and the retainer on the tube. Install the capscrews.



Torque Value: 35 n•m [25 ft-lb]

Tighten and torque the 24 cover capscrews in the steps and values listed below.

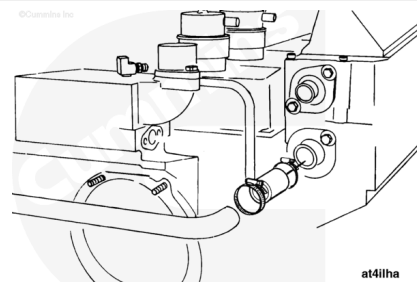
Torque Value:	Step 1	35 n.m [25 ft-lb]
	Step 2	50 n.m [35 ft-lb]



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Install the inlet hose and clamps. Tighten the clamps.

Torque Value: 6 n•m [50 in-lb]

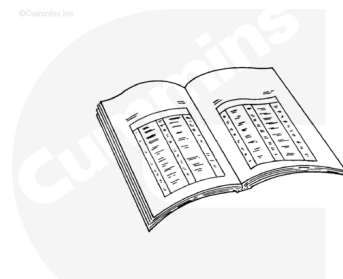


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Finishing Steps

⚠ CAUTION ⚠

Use caution when filling the cooling system that the coolant is not spilled into the bilge area. Do not pump coolant overboard. If the coolant is spilled, it must be disposed of in accordance with local environmental regulations



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- Marine applications: Install the aftercooler coolant tubes. Refer to Procedure 010-006 (</qs3/pubsys2/xml/en/procedures/20/20-010-006.html>)
- Install the aftercooler coolant vent tube. Refer to Procedure 008-017 (</qs3/pubsys2/xml/en/procedures/20/20-008-017.html>).
- Install the air crossover tube. Refer to Procedure 010-019 (</qs3/pubsys2/xml/en/procedures/20/20-010-019-tr.html>).
- Fill the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>).
- Operate the engine and check for leaks.

